MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-716/A, HARNESS ASSEMBLY

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the harness assembly described herein shall consist of this document and the latest issue in effect of Specification MIL-B-49450(ER).

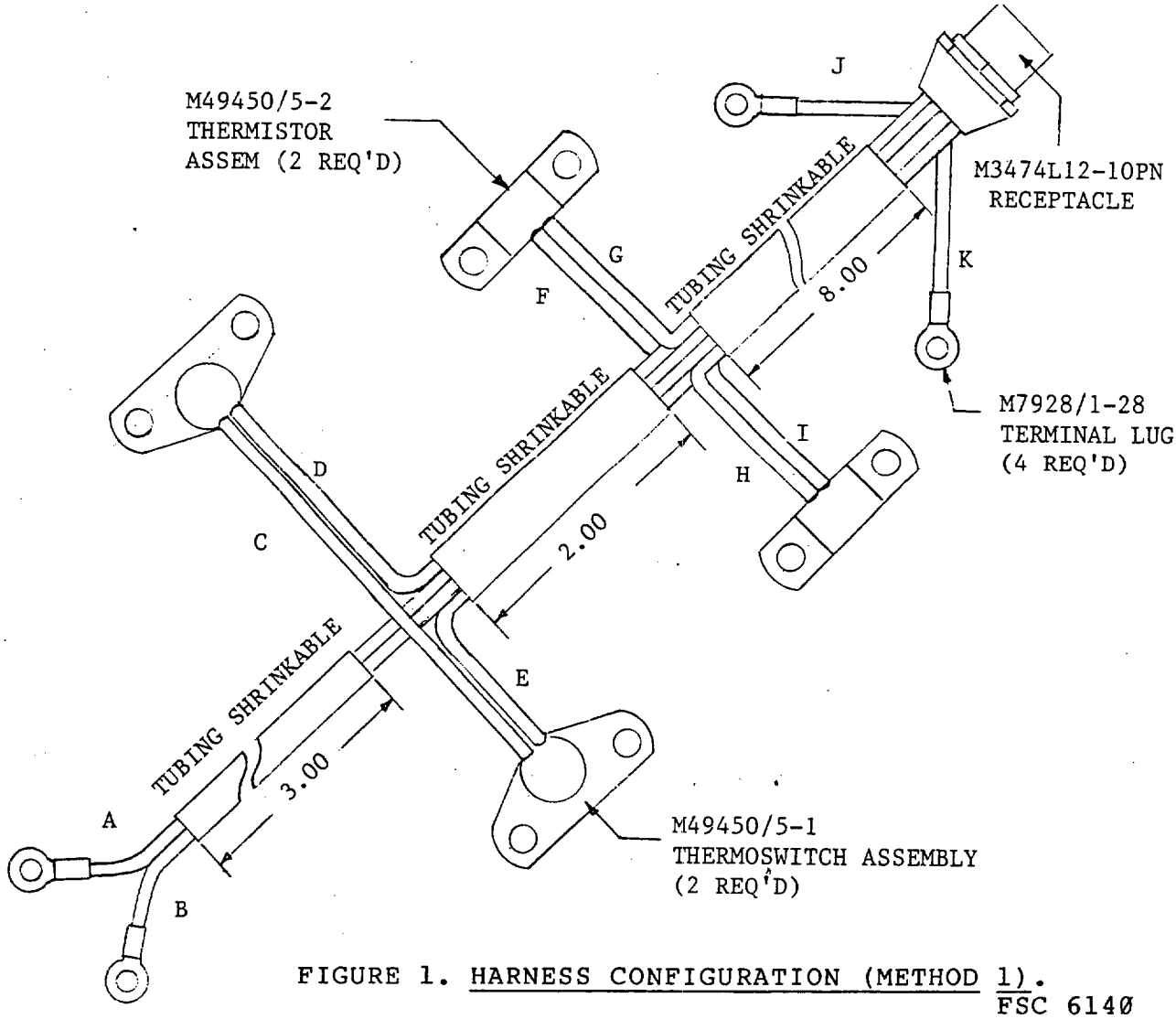


FIGURE 1. HARNESS CONFIGURATION (METHOD 1).

FSC 6140

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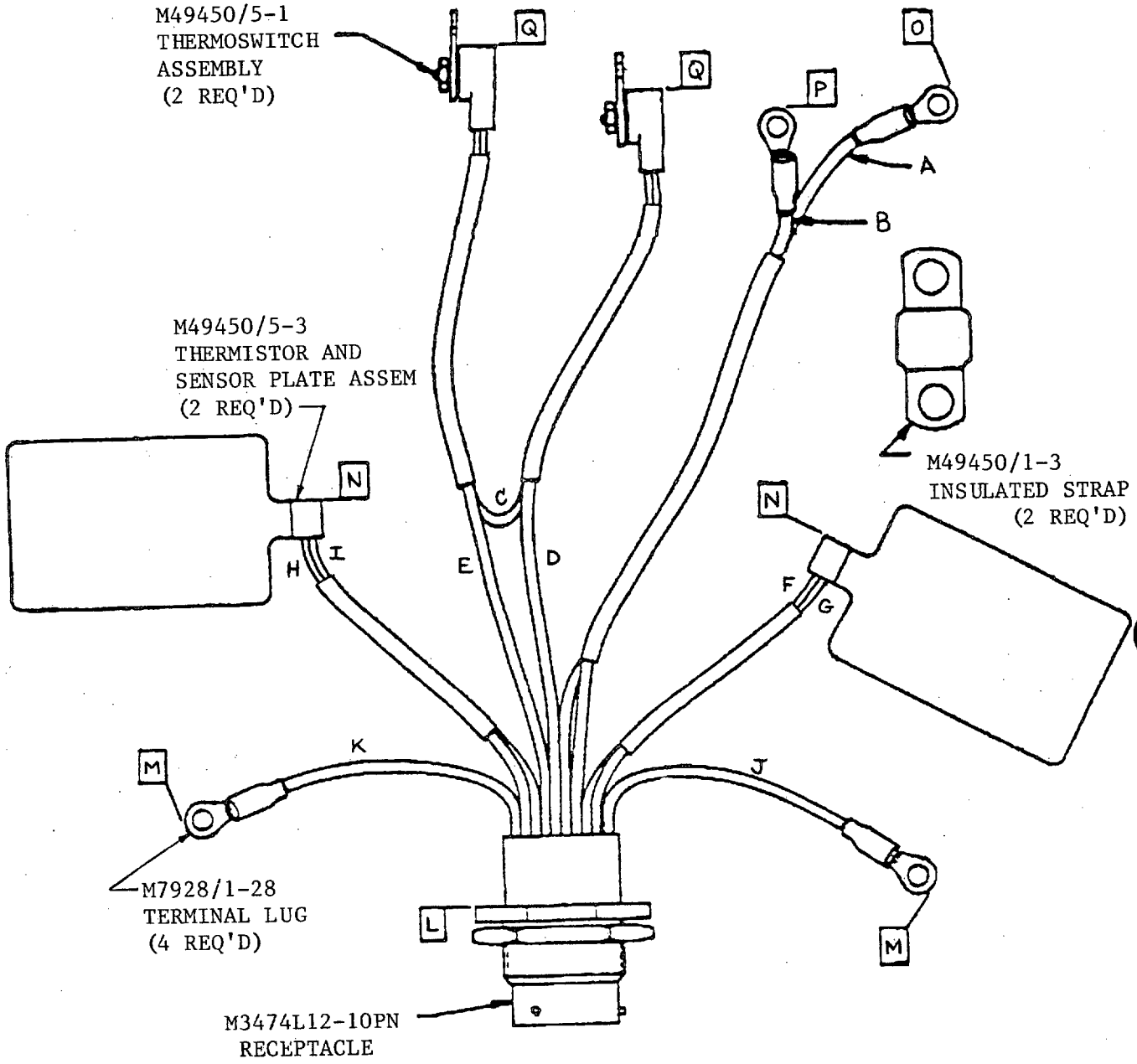


FIGURE 2. HARNESS CONFIGURATION (METHOD 2).

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TABLE 1. REFERENCE WIRE LISTING FOR FIGURE 1.

REF LTR	WIRE DESCRIPTION
A	WIRE, BLUE TO PIN C, 18 INCHES LONG
B	WIRE, GREEN TO PIN H, 16.5 INCHES LONG
C	WIRE, WHITE, 2.5 INCHES LONG
D	WIRE, WHITE TO PIN E, 14 INCHES LONG
E	WIRE, WHITE TO PIN F, 12 INCHES LONG
F	WIRE, YELLOW TO PIN A, 10 INCHES LONG
G	WIRE, YELLOW TO PIN B, 10 INCHES LONG
H	WIRE, ORANGE TO PIN K, 10 INCHES LONG
I	WIRE, ORANGE TO PIN J, 10 INCHES LONG
J	WIRE, RED TO PIN G, 6 INCHES LONG
K	WIRE, BLACK TO PIN D, 6 INCHES LONG

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Table 2. Reference Wire Listing for Figure 2.

REF LTR	WIRE DESCRIPTION
A	WIRE, BLUE TO PIN C, 16.0 ± .50 INCHES LONG FROM L TO O
B	WIRE, GREEN TO PIN H, 15.3 ± .50 INCHES LONG FROM L TO P
C	WIRE, WHITE AS REQUIRED
D	WIRE, WHITE TO PIN E, 14.38 ± .50 INCHES LONG FROM L TO Q
E	WIRE, WHITE TO PIN F, 14.38 ± .50 INCHES LING FROM L TO Q
F	WIRE, YELLOW TO PIN A, 9.12 ± .50 INCHES LONG FROM L TO N
G	WIRE, YELLOW TO PIN B, 9.12 ± .50 INCHES LONG FROM L TO N
H	WIRE, ORANGE TO PIN K, 9.12 ± .50 INCHES LONG FROM L TO N
I	WIRE, ORANGE TO PIN J, 9.12 ± .50 INCHES LONG FROM L TO N
J	WIRE, RED TO PIN G, 6.50 ± .50 INCHES LONG FROM L TO M
K	WIRE, BLACK TO PIN D, 6.50 ± .50 INCHES LONG FROM L TO M

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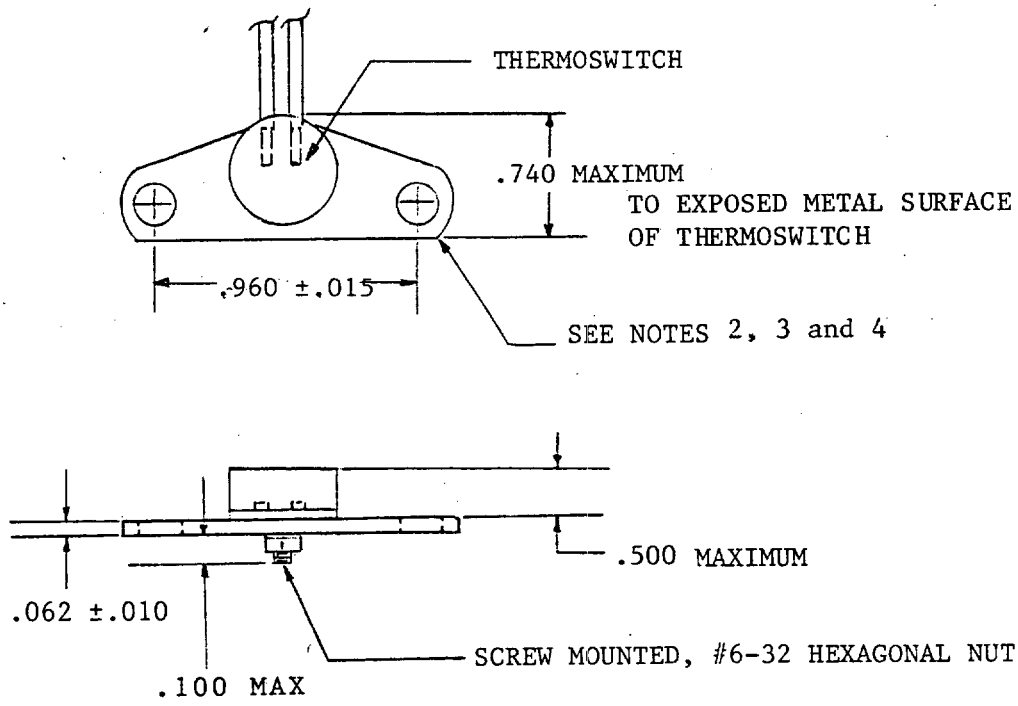


FIGURE 3. M49450/5-1 THERMOSWITCH ASSEMBLY.

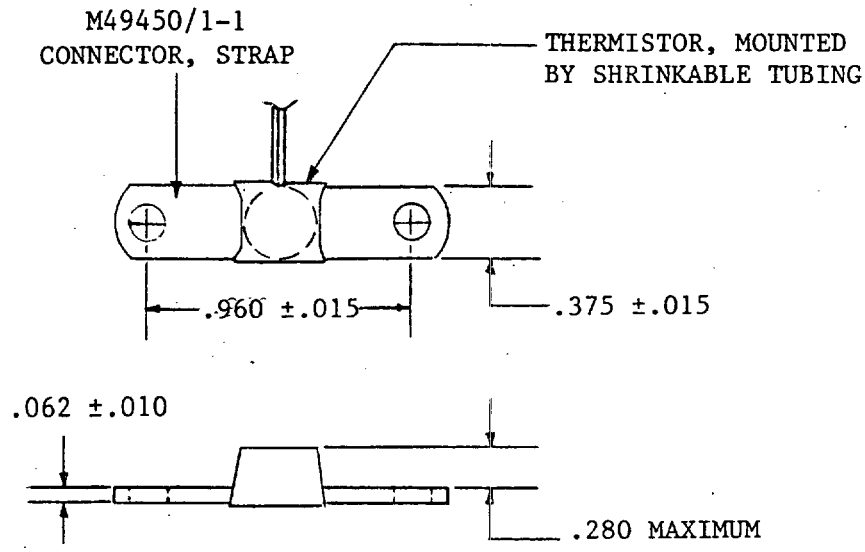


FIGURE 4. M49450/5-3 THERMISTOR ASSEMBLY (METHOD 1).

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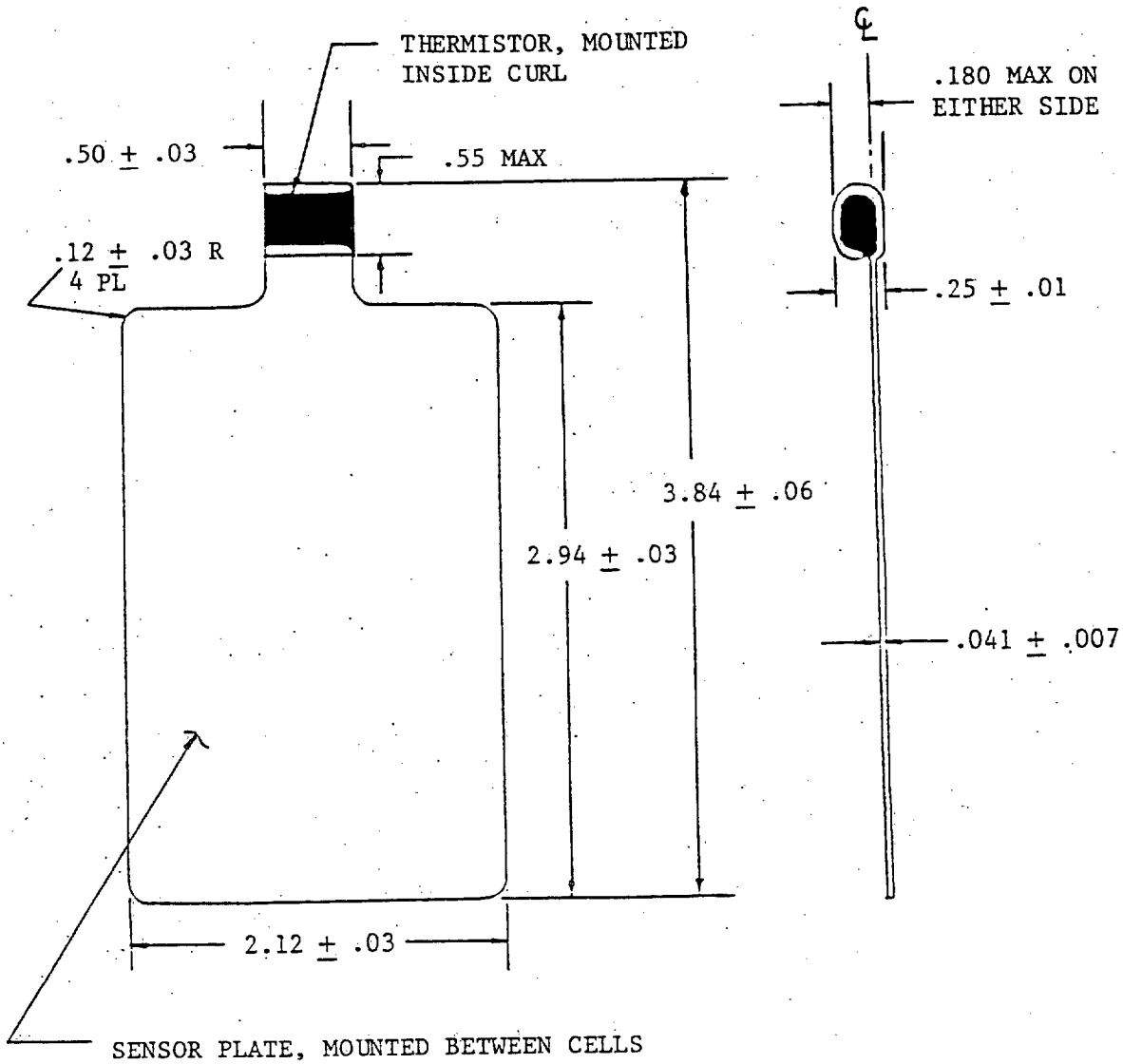
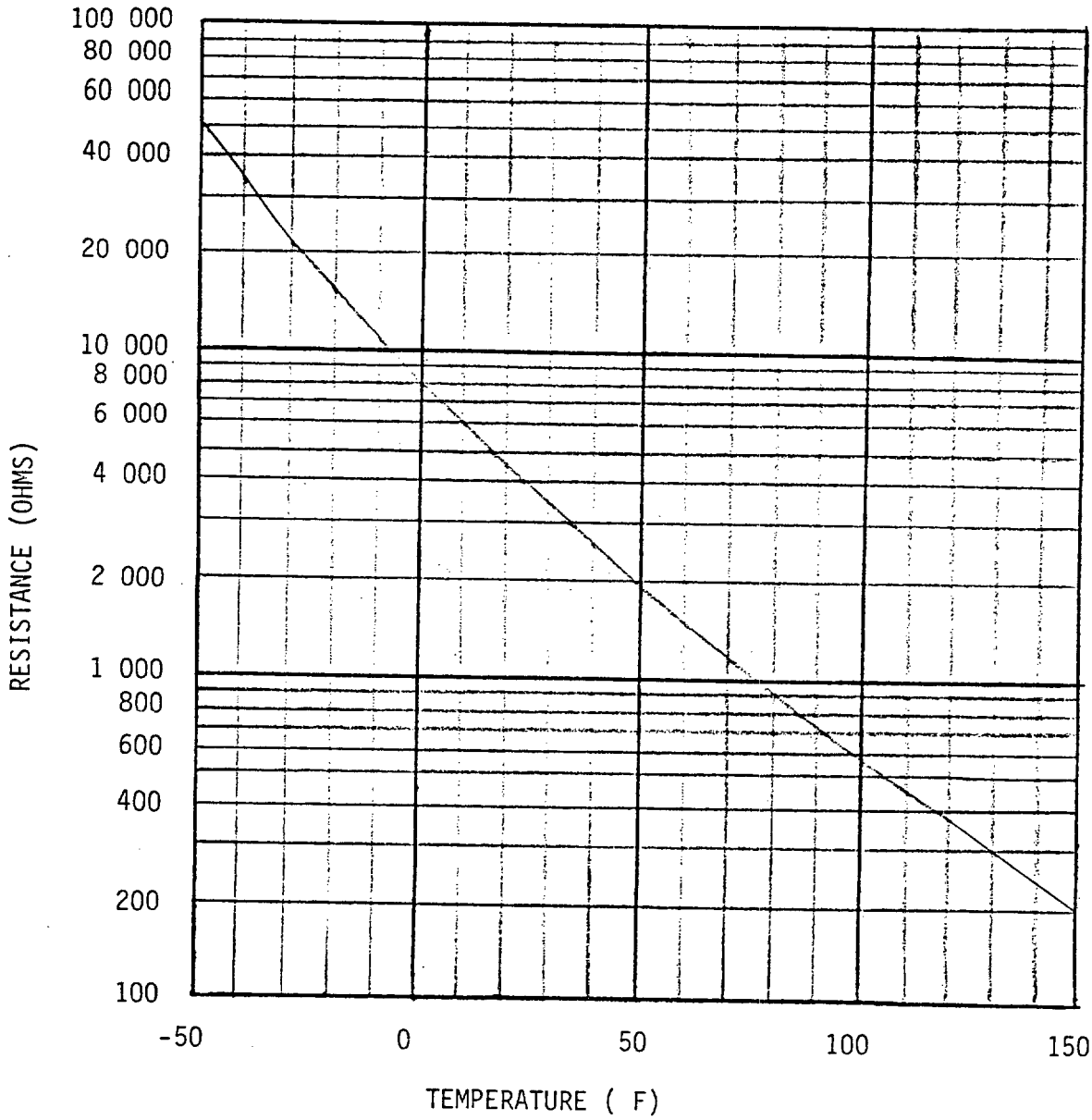


FIGURE 5. M49450/5-3 THERMISTOR ASSEMBLY WITH SENSOR PLATE (METHOD).

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FIGURE 6. THERMISTOR OR TEMPERATURE DEVICE RESISTANCE-TEMPERATURE CHARACTERISTICS.



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DRAWING NOTES:

1. All dimensions are in inches, unless otherwise specified.

2. Strap shall be constructed of copper alloy conforming to QQ-C-576 or QQ-C-502.

3. Strap shall be plated in accordance with QQ-N-290, Class I, Grade E.

4. Strap shape is optional.

REQUIREMENTS:

1. The harness assembly shall use 20 AWG, teflon insulated wires conforming to MIL-W-16978 and resistant to the battery electrolyte. The length and color of wires shall be in accordance with Figure 1 and Table 1 or Figure 2 and Table 2.

2. All electrical connections made to the MS3474-L12-10PN receptacle and harness devices (thermistors, thermoswitches or equal) shall be potted with a compound that will not support flame, absorb moisture, or be damaged by battery electrolyte. Thermoswitch shall be modified to achieve maximum height of Figure 3.

3. The thermistor or temperature sensor device shall exhibit the resistance temperature characteristics of Figure 6.

4. The harness assembly shall be capable of withstanding the temperature and exposure of the battery electrolyte without damage.

Preparing Activity:
Army-ER

(Project 6140-A601-05)

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